

VAIDIK KUMAR

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PROFILE SUMMARY

Computer Science undergraduate with expertise in Python, Data Structures & Algorithms, Machine Learning, and Generative AI technologies. Hands-on experience in developing intelligent systems, including a reinforcement learning-based chess engine and a CNN-based malaria detection model. Proficient in building scalable applications using modern AI frameworks. Strong analytical, problem-solving, and teamwork skills demonstrated through hackathons and technical competitions.

EXPERIENCE

Oni Infotech

May 2025 – July 2025

Machine Learning Intern

- Developed and evaluated machine learning models for detecting phishing emails, leveraging NLP-based features to identify malicious intent.
- Engineered a robust feature extraction pipeline using TF-IDF and n-grams to analyze email body text, headers, and sender/domain patterns with Python and Scikit-learn.
- Trained and compared Logistic Regression, Random Forest, and SVM models using precision, recall, and F1-score to determine the most effective classifier.
- Managed end-to-end data preprocessing workflows, including text cleaning, handling class imbalance via oversampling, and ensuring model stability through cross-validation.
- Analyzed URL patterns and domain reputation to improve detection accuracy and reduce false positive rates in real-time classification scenarios.

PROJECTS

Malaria Detection with Deep Learning

Aug 2025 – Dec 2025

Tech Stack: Python, TensorFlow, OpenCV, NumPy

- Developed and optimized a CNN (EfficientNetB3) to classify blood smear images as parasitized or uninfected, utilizing OpenCV for advanced image preprocessing and noise reduction.
- Implemented comprehensive data augmentation strategies and hyperparameter tuning to mitigate overfitting, ultimately achieving 94% classification accuracy for potential clinical application.

Reinforcement Learning vs Large Language Model in Chess Engine

Jan 2026 – Apr 2026

Tech Stack: Python, PyTorch, PPO, Transformers, OpenAI Gym

- Developed a chess-playing agent using Proximal Policy Optimization (PPO) and compared it with LLM-based move generation for strategic decision-making.
- Integrated OpenAI Gym environments with custom chess logic to simulate and train agents against established open-source engines.
- Conducted a comparative analysis highlighting key differences in reward shaping, state representation, and long-term planning between RL and LLM approaches.

SKILLS

- Languages:** C, C++, Python, Java, JavaScript, SQL
- AI/ML:** Machine Learning, NLP, LLMs, RAG, Computer Vision, Agentic AI
- Frameworks/Libraries:** Flask, Django, LangChain, LangGraph, scikit-learn, Pandas, NumPy, Matplotlib
- Tools & Databases:** Git, GitHub, Jupyter Notebook, VS Code, MySQL, ChromaDB, Cursor
- Core Concepts:** DSA, OOP, DBMS, OS

ACHIEVEMENTS

- Technovation Hackathon 2025** – Qualified for finals among 50 collegiate teams by designing a real-time adaptive traffic signal system to optimize congestion and reduce fuel emissions.
- Elementary SciTech 2024** – Runner-Up in the Open SciTech Contest at IIT Delhi (AI & Robotics).
- Competitive Programming** – Solved 250+ LeetCode problems; 1000+ rating on Codeforces.

CERTIFICATIONS & COURSEWORK

- Oracle Cloud Infrastructure 2025 Certified AI Foundations:** AI fundamentals, ML concepts, and OCI AI services
- Machine Learning by Andrew Ng (Coursera):** Covered Supervised & Unsupervised Learning
- Corporate Social Responsibility (NPTEL - ONT502):** Learned CSR fundamentals and implementation.

EDUCATION

Sharda University

Aug 2023 – Present

Bachelor of Technology in Computer Science